BASIC CELLULAR SYSTEM

There are mainly two types of Basic Cellular System:

1. Circuit Switched:
In a circuit-switched system, each traffic channel is dedicated to a user until its cell is terminated.

2. Packet Switched:
In packet-switching, the packets are sent towards the destination irrespective of each other. Each packet has to find its own route to the destination. There is no predetermined path; the decision as to which node to hop to in the next step is taken only when a node is reached. Each packet finds its way using the information it carries, such as the source and destination IP addresses.

COMPONENTS OF A BASIC CELLULAR SYSTEM

Cell
- Covers geographical region.

Base Station (BS)
- Analogous to 802.11 AP.

Mobile Users
- Attach to network through BS.

Air-interface
- Physical and link layer protocol between mobile and BS.

MSC (Mobile Switching Center)
- Connects cells to wired telephone network.
- Manages cell setup.
- Handles mobility.



Mobile switching Center

Mobile switching Center

PUBLIC TELEPHONE SYSTEM

Wired Network

Analog Circuit- Switched System

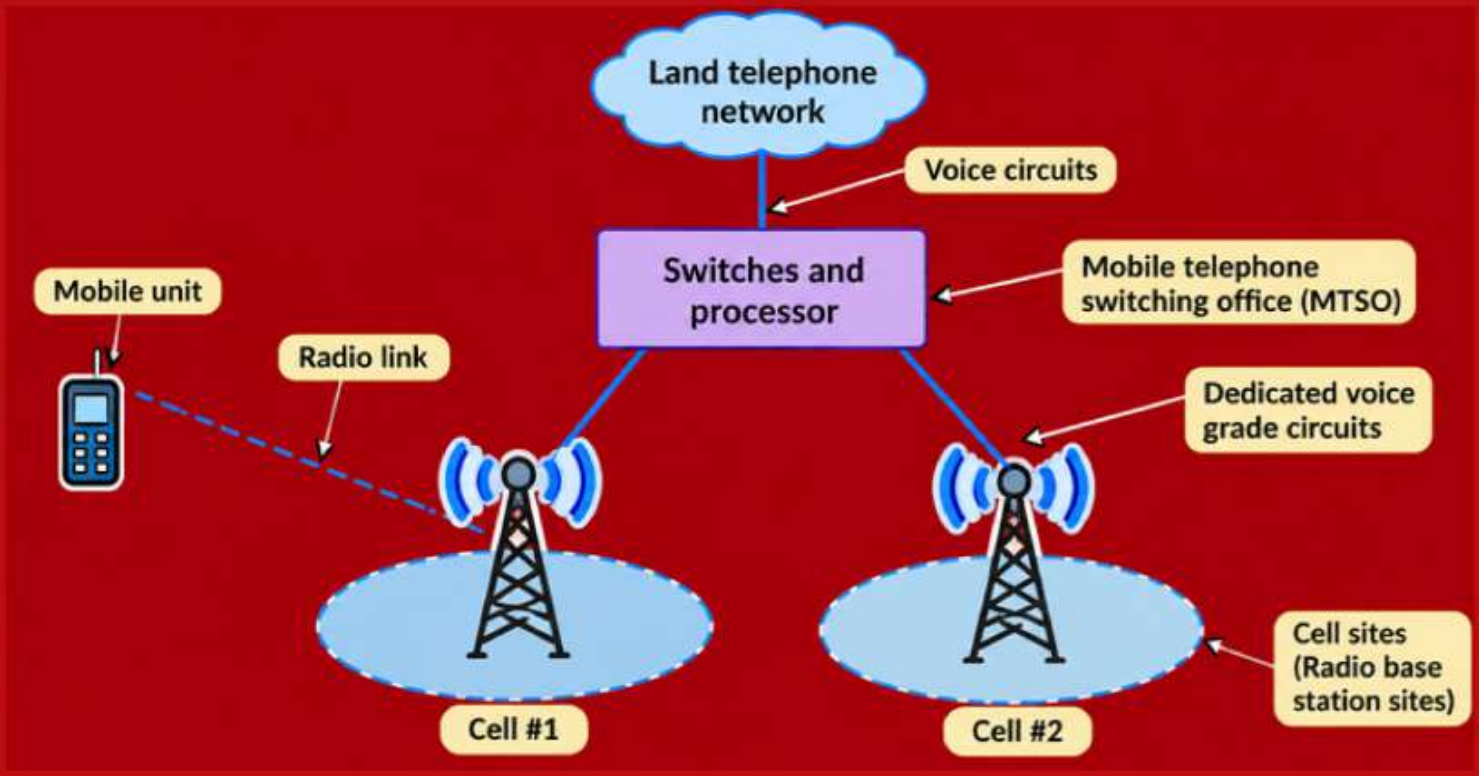
A basic analog cellular system consists of three subsystems:

1. A Mobile Unit

2. A Cell Site

3. A Mobile Telephone Switching Office (MTSO) with CONNECTIONS to link the three subsystems.

Analog Circuit-Switched System

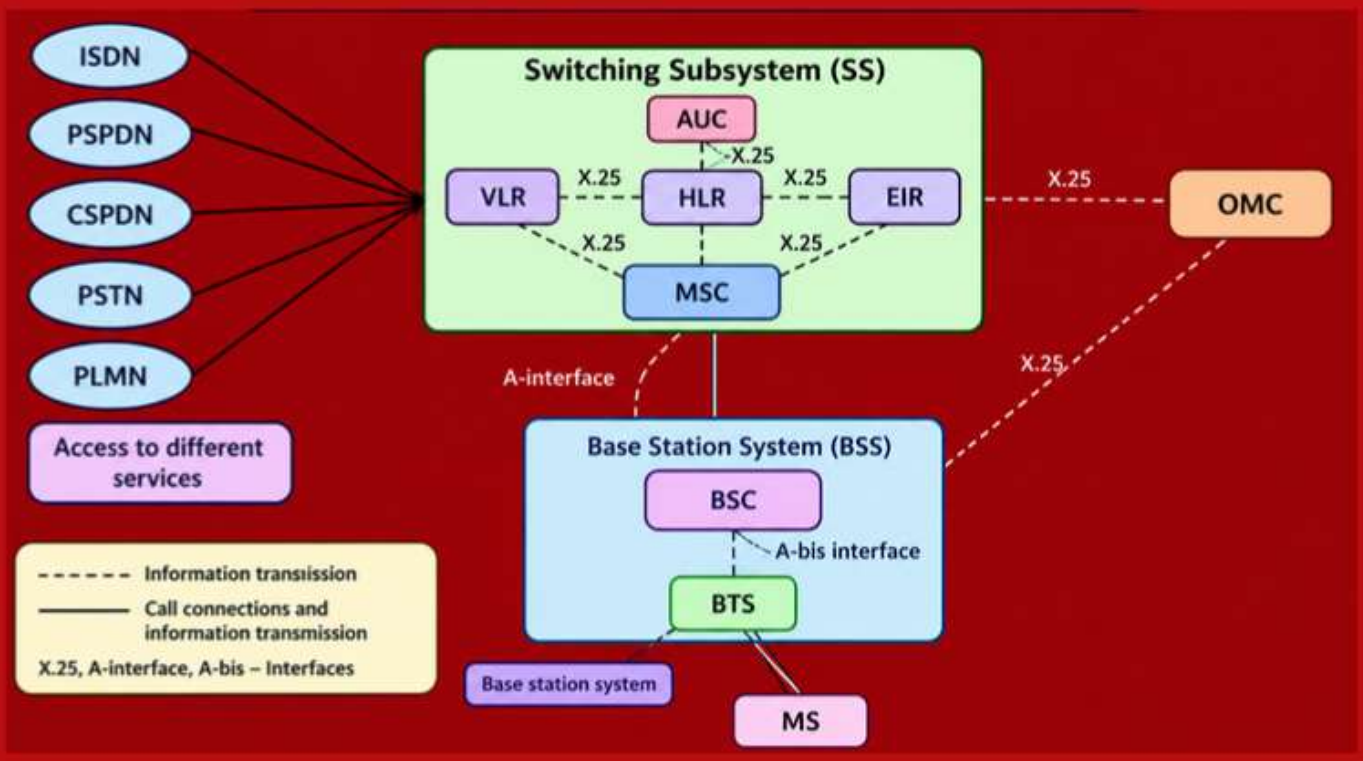


Digital Circuit-Switched System

A Basic Digital System consists of four elements:

1. Mobile Station
2. Base Transceiver Station (BTS)
3. Base Station Controller (BSC)
4. Switching Subsystems

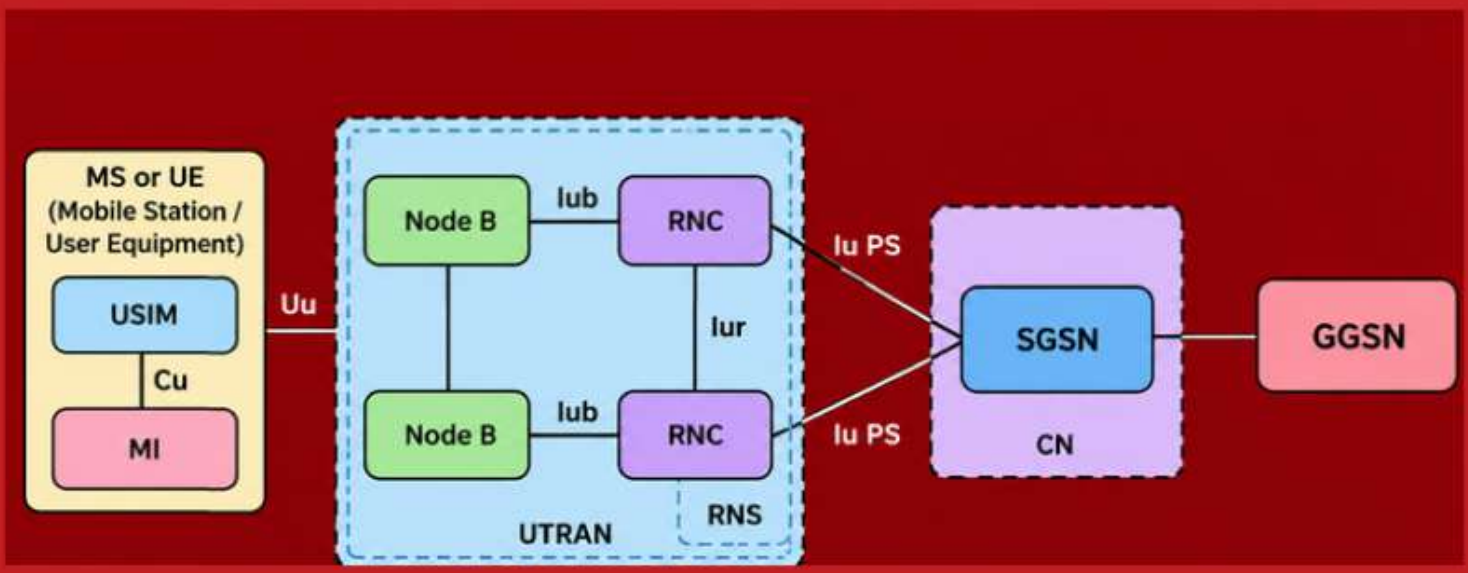
Digital Circuit-Switched System



Packet Switched System

A cellular packet-switched system has six elements as follows:

1. MS (Mobile Station)
2. Node B
3. RNC (Radio Network Controller)
4. SGSN (Service GPRS Support Node)
5. GGSN (Gateway GPRS Support Node)
6. CGF (Changing Gateway Function)



Information transmission

Call connections and information transmission

CELLULAR PACKET SYSTEMS